

Business Problem Statement

A leading retail company wants to better understand its customers' shopping behavior to improve sales, increase customer satisfaction, and build long-term loyalty. The management team has observed that purchasing patterns are changing across different customer groups, product categories, and sales channels, including both online and offline stores.

The company wants to identify which factors influence customer purchase decisions and encourage repeat purchases. These factors include:

- **Discounts:** Are promotions or price reductions motivating purchases?
- **Product Reviews:** Do positive reviews increase the likelihood of buying a product?
- **Seasonal Trends:** Are customers buying certain products more during specific seasons or festivals?
- **Payment Preferences:** Do different payment options affect purchase decisions?

The goal is to clearly understand how each of these factors impacts customer behavior, so the company can understand customers better, keep them interested, and encourage them to buy again.

Business Goal

The goal of this project is to help the company leverage customer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies.

Key Question

"How can the company use customer shopping data to identify patterns, improve customer experience, and plan better marketing and product strategies?"

Deliverables

1. **Data Preparation & Modeling (Python):**
Clean and transform the raw dataset to make it ready for analysis.
2. **Data Analysis (SQL):**
Organize the data into a structured format, simulate business transactions, and run queries to extract insights on customer segments, loyalty, and factors driving purchases.
3. **Visualization & Insights (Power BI):**
Build an interactive dashboard that highlights key patterns and trends, enabling stakeholders to make informed decisions.
4. **Report & Presentation:**
Summarize the key findings and business recommendations in a clear report, and prepare a presentation that communicates insights visually and effectively to stakeholders.
5. **GitHub Repository:**
Include all Python scripts, SQL queries, and dashboard files in a well-structured repository to ensure reproducibility and transparency.